

The role of entropy in Beamer themes

A template that stays out of the way

Your Name — Quantum Thermodynamics Group Basel

27.08.2026



Outline

- ▶ Motivation

- ▶ Building a slide

- ▶ Conclusion

Solid-state qubits

THE ENVIRONMENT SETS THE COHERENCE TIME

- $N = 10^4 - 10^5$ nuclear spin bath
- Huge pure dephasing, $T_2^* = 3.9$ ns
- Bath engineering enhances qubit coherence



N. G. Nguyen et al. *Enhanced Electron-Spin Coherence in a GaAs Quantum Emitter*, Phys. Rev. Lett. **131**, 210805 (2023)

D. M. Jackson et al. *Optimal Purification of a Spin Ensemble by Quantum-Algorithmic Feedback*, Phys. Rev. X **12**, 031014 (2022)

Outline

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One idea, revealed one step at a time

PROGRESSIVE BUILD

Duplicate the slide, add one element, keep the title. Three devices mark what is new, and nothing else:

- A red underline under the term being discussed

This line is greyed out until the third overlay.

One idea, revealed one step at a time

PROGRESSIVE BUILD

Duplicate the slide, add one element, keep the title. Three devices mark what is new, and nothing else:

- A red underline under the term being discussed
- A thin red box around the statement that just landed

Aligned $m \rightarrow m + 1$, anti-aligned $m \rightarrow m - 1$

This line is greyed out until the third overlay.

One idea, revealed one step at a time

PROGRESSIVE BUILD

Duplicate the slide, add one element, keep the title. Three devices mark what is new, and nothing else:

- A red underline under the term being discussed
- A thin red box around the statement that just landed
- Greying out what is not yet in play

Aligned $m \rightarrow m + 1$, anti-aligned $m \rightarrow m - 1$

This line is greyed out until the third overlay.

The equation is the anchor

MASTER EQUATION

A content slide carries one anchor — an equation, a figure or a schematic — never two.

$$\partial_t \rho_m = -i [\underline{\mathbf{B}_m} \cdot \mathbf{S}, \rho_m] + \Gamma_m^+ \mathcal{D}[\sigma^\dagger] \rho_m + \Gamma_m^- \mathcal{D}[\sigma] \rho_m$$

- $\rho_S = \sum_m \rho_m$, with $\text{tr } \rho_m = P_m$
- Symbols are defined before they are used

Blocks, alerts and links

A block

Quiet by design: a tinted panel with a petrol heading, never a full-width coloured bar.

An alert block

Basel red is the highlight colour. One emphasis per slide wins.

Citations can also be footnotes.¹

- Alert text is Basel red: **important**
- Links are petrol: unibas.ch
- Blocks stay tinted, never barred

¹N. G. Nguyen et al., “Enhanced electron-spin coherence in a GaAs quantum emitter”, *Phys. Rev. Lett.* **131**, 210805 (2023).

A chapter slide

The chapter environment fills the slide with one colour and clips an image into the right-hand wedge.



A slide with a side picture

- The sidepic environment bleeds an image off the right edge
- Text keeps the left 55 % of the slide



Outline

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Conclusion

- One idea per slide, one anchor, at most four bullets
- Red marks what is new; petrol carries the structure
- Every borrowed result gets its reference in the corner
- Backup slides live after the closing slide

Thanks!
Questions?

TEMPLATE AT [GITHUB.COM/MARCELOJBP/UNIBASELBEAMERTEMPLATE](https://github.com/marcelojbp/unibase-beamer-template)

Backup: the colour palette

#	Name	Macro	Use
1	Mint	unibasMint	Title band, dividers
2	Turquoise	unibasTurquoise	Secondary accent
3	Anthracite	unibasAnthracite	Body text
5	Basel red	unibasRed	Emphasis, data
8	Petrol	unibasPetrol	Structure, model

References

- D. M. Jackson et al., “Optimal purification of a spin ensemble by quantum-algorithmic feedback”, *Phys. Rev. X* **12**, 031014 (2022).
- N. G. Nguyen et al., “Enhanced electron-spin coherence in a GaAs quantum emitter”, *Phys. Rev. Lett.* **131**, 210805 (2023).
- A. Riera-Campeny, A. Sanpera, and P. Strasberg, “Quantum systems correlated with a finite bath: nonequilibrium dynamics and thermodynamics”, *PRX Quantum* **2**, 010340 (2021).